

Statistical and Machine Learning

Exemplar and kernel methods

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Exemplar and kernel methods

- ▶ Exemplar methods
 - ▶ k-nearest neighbor (kNN)
 - ▶ kernel density estimation (KDE)
- ▶ Kernel methods
 - ▶ Mercer Kernels
 - ▶ Gaussian process (GP)
 - ▶ Support Vector Machine (SVM)

Exemplar methods vs kernel methods

- ▶ **Exemplar methods** predict using stored training examples.
- ▶ They are often **local**: nearby observations matter most.
- ▶ Examples: **k -NN**, **kernel density estimation**.

- ▶ **Kernel methods** use a kernel $K(\mathbf{x}, \mathbf{z})$ as an implicit **similarity / inner-product function**.
- ▶ They often rely on the **kernel trick**.
- ▶ Examples: **SVM**, **kernel ridge regression**, **Gaussian processes**.

Key difference

Exemplar methods use stored examples directly, often with local distance-based weighting.

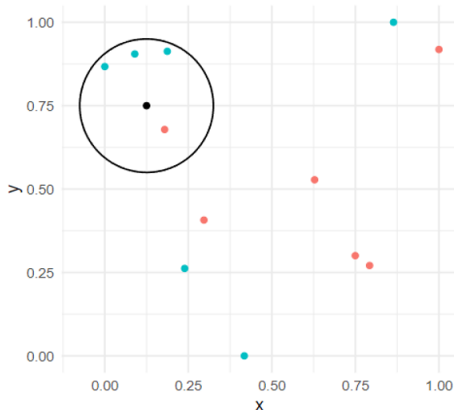
Kernel methods use positive semidefinite kernels as implicit inner products in feature space.

Exemplar and kernel methods

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k-nearest neighbor – basic idea

- ▶ Points that are similar in the predictor space are expected to have similar responses.
- ▶ Given a new point (black), which class do you think it is more likely to belong to?



k-nearest neighbor – classification rule

- ▶ Consider a classification setting.
- ▶ The k -NN model estimates the probability that an observation with features $\mathbf{x}$ belongs to class l based on the labels of the k nearest neighbors of $\mathbf{x}$, denoted by $N_{\mathbf{x}}^k$.

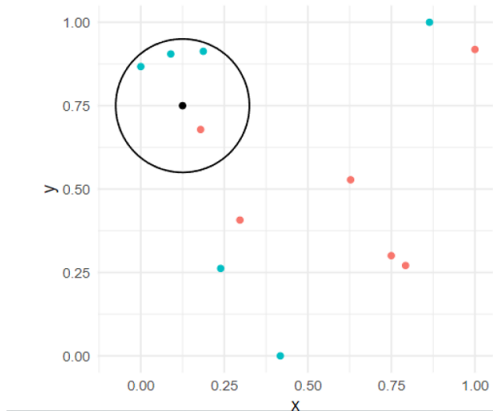
$$\hat{P}(Y = l \mid \mathbf{x}) = \frac{1}{k} \sum_{i \in N_{\mathbf{x}}^k} 1_{\{y_i=l\}}$$

- ▶ The term $\sum_{i \in N_{\mathbf{x}}^k} 1_{\{y_i=l\}}$ counts how many of the k nearest neighbors belong to class l .
- ▶ Dividing by k gives the estimated probability.

k-nearest neighbor – example

Suppose $k = 4$ is chosen. At the candidate black point. The four nearest neighbors are inspected.

There is a probability of $\frac{3}{4}$ of being in the green group and $\frac{1}{4}$ for being in the orange group.



k-nearest neighbor – regression

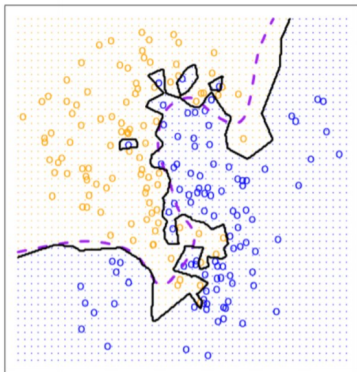
- ▶ Consider a regression setting.
- ▶ The k -NN model predicts the response value for an observation with features $\mathbf{x}$ based on the response values of the k nearest neighbors of $\mathbf{x}$, denoted by $N_{\mathbf{x}}^k$.

$$\hat{f}(\mathbf{x}) = \hat{\mathbb{E}}(Y \mid \mathbf{x}) = \frac{1}{k} \sum_{i \in N_{\mathbf{x}}^k} y_i$$

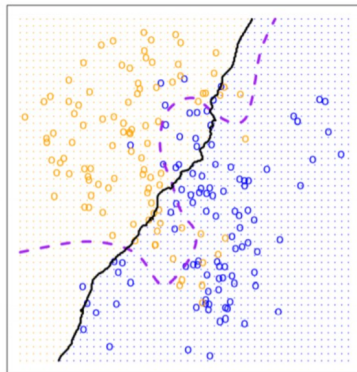
- ▶ The term $\sum_{i \in N_{\mathbf{x}}^k} y_i$ adds the response values of the k nearest neighbors.
- ▶ Dividing by k gives the average response among the nearest neighbors, which is the k -NN regression prediction.

k-nearest neighbor – different choices of k

KNN: $K=1$



KNN: $K=100$



How do we choose $N_{\mathbf{x}}^k$?

- ▶ $N_{\mathbf{x}}^k$ depends on the choice of distance metric.
- ▶ Euclidean distance:

$$L_2(\mathbf{x}_i, \mathbf{x}_j) = \left(\sum_{l=1}^p |x_{il} - x_{jl}|^2 \right)^{1/2}$$

- ▶ More generally, the L_q distance is

$$L_q(\mathbf{x}_i, \mathbf{x}_j) = \left(\sum_{l=1}^q |x_{il} - x_{jl}|^q \right)^{1/q}$$

- ▶ Manhattan distance is the special case $q = 1$:

$$L_1(\mathbf{x}_i, \mathbf{x}_j) = \sum_{l=1}^q |x_{il} - x_{jl}|$$

- ▶ The set $N_{\mathbf{x}}^k$ contains the k training points closest to $\mathbf{x}$ under the chosen distance.

kNN vs. LDA vs. Logistic Regression

- ▶ k -NN takes a completely different approach.
- ▶ k -NN is fully nonparametric: it makes no assumptions about the shape of the decision boundary.
- ▶ Advantage of k -NN: it can outperform both LDA and logistic regression when the decision boundary is highly nonlinear.
- ▶ Disadvantage of k -NN: it does not indicate which predictors are important, since there is no table of coefficients.

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Density estimation

- ▶ In exploratory data analysis, density estimation can be used:
 - ▶ to assess multimodality, skewness, tail behavior, etc.,
 - ▶ in decision-making, classification, and the summarization of Bayesian posteriors,
 - ▶ as a useful visualization tool, providing a simple summary of a distribution.
- ▶ Suppose random variables $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ have been observed and are assumed to be sampled independently from a distribution with density f .
- ▶ Goal: to estimate the density function f .

Density estimation

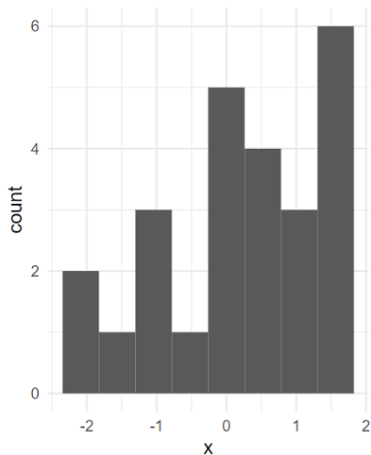
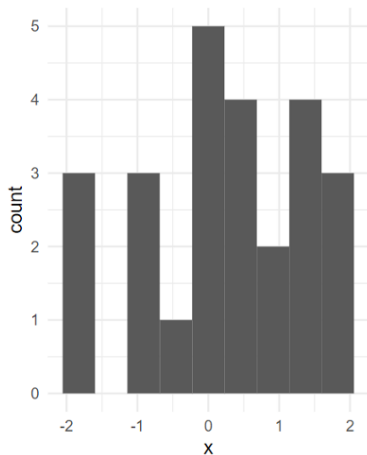
- ▶ Parametric method: assume a parametric model, e.g., assume that your data follows a normal distribution:

$$\mathbf{X}_1, \dots, \mathbf{X}_n \sim f_{\theta}(\mathbf{x}), i.i.d.$$
$$\theta = (\boldsymbol{\mu}, \boldsymbol{\Sigma}) \text{ and } \mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

- ▶ We can use MLE to estimate parameters
- ▶ What if the data do not follow parametric assumption?
 - ▶ Histogram!

Histogram

- ▶ Crude density estimate (kinky edges of bins)
- ▶ Offers a very simple visualization
- ▶ Sensitive to the number of bins chosen and bin width

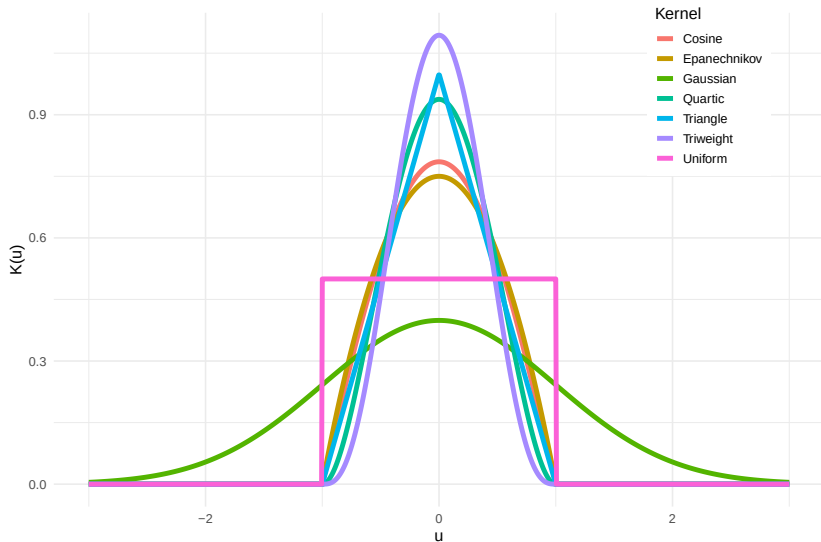


Kernel density estimation

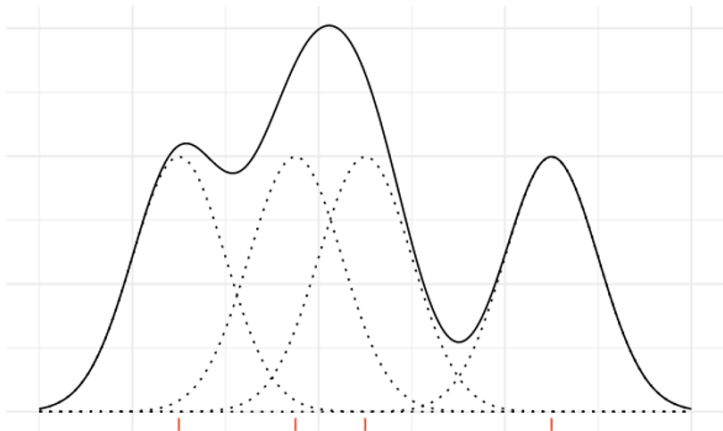
- ▶ Kernel density estimation is a nonparametric approach to density estimation.
- ▶ At each data point, a kernel function is centered at that observation.
- ▶ The density is estimated by summing these kernel functions and dividing by the number of observations, so that the estimator satisfies:
 - ▶ non-negativity,
 - ▶ the integral over its support equals 1.

Kernel functions

A kernel $K(\mathbf{x})$ is a density function with following properties: a) non-negative $K(\mathbf{x}) \geq 0$, b) symmetric $K(-\mathbf{x}) = K(\mathbf{x})$, c) unit measure $\int K(\mathbf{x})d\mathbf{x} = 1$.



Normal kernel density estimate



Example: Four sampled variables marked in red with Gaussian weights sum together to give the overall density estimate.

Kernel density estimator (KDE)

- ▶ A simple estimator weights all points within a window of width $2h$ around x equally:

$$\hat{f}(x) = \frac{1}{2nh} \sum_{i=1}^n 1_{\{|X_i - x| \leq h\}}$$

- ▶ More generally, a univariate kernel density estimator uses a kernel function K to assign weights:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)$$

where:

- ▶ K is a kernel function
- ▶ h is a bandwidth parameter, which may be fixed or vary with x

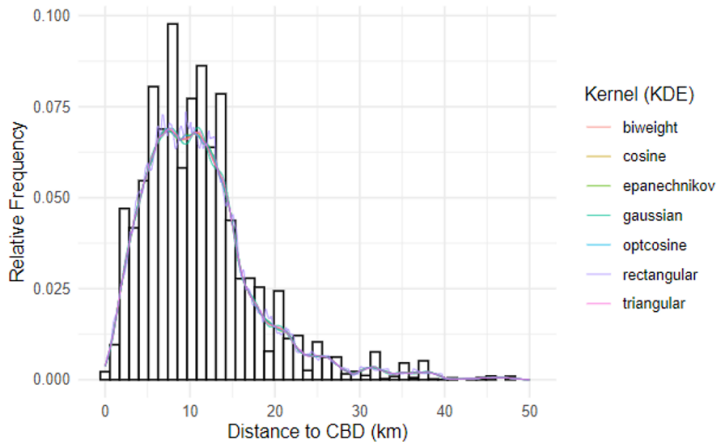
Tuning the kernel density estimator (KDE)

- ▶ The choice of kernel K is usually less important, as different kernels often produce similar results.
- ▶ The choice of bandwidth h is crucial, as it controls the degree of smoothing.

Some widely used kernels:

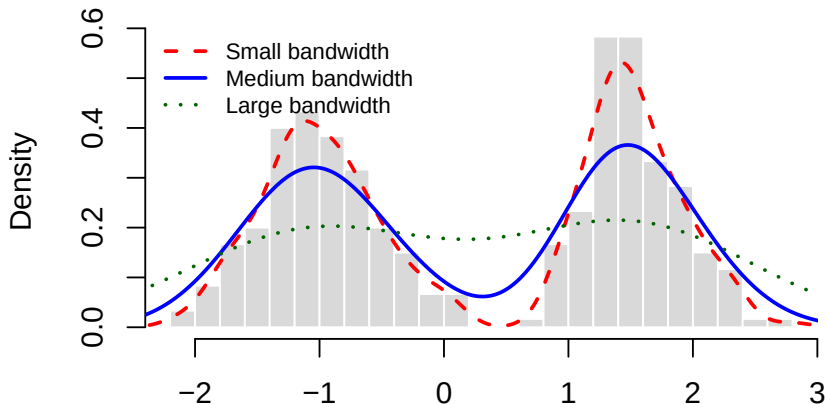
- ▶ Uniform: $K(x) = \frac{1}{2}1_{\{|x| \leq 1\}}$
- ▶ Gaussian: $K(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\}$
- ▶ Epanechnikov: $K(x) = \frac{3}{4}(1 - x^2)1_{\{|x| \leq 1\}}$

Different choices of kernel function with the same bandwidth



Choice of bandwidth

- ▶ If h is too small, the density estimator assigns probability density too locally near the observed data → a *wiggly estimated density with many spurious modes*.
- ▶ If h is too large, the density estimator spreads probability density contributions too diffusely → important features of f may be smoothed away,



Bandwidth selection by leave-one-out (LOO) cross-validation

The bandwidth h controls the smoothness of the KDE. A common criterion is based on the integrated squared error:

$$R(h) = \int \{\hat{p}_h(x) - p(x)\}^2 dx.$$

Since the true density p is unknown, we minimize an empirical version of this risk. Ignoring the constant term $\int p^2(x)dx$, the leave-one-out cross-validation criterion is

$$\hat{R}_{\text{LOO}}(h) = \int \hat{p}_h^2(x)dx - \frac{2}{n} \sum_{i=1}^n \hat{p}_{h,-i}(X_i),$$

where $\hat{p}_{h,-i}$ is the KDE computed after removing observation X_i .

$$\hat{h}_{\text{LOO}} = \arg \min_{h \in \mathcal{H}} \hat{R}_{\text{LOO}}(h).$$

Idea: evaluate how well the density estimated from all observations except X_i predicts the omitted observation X_i . This avoids evaluating the estimator on the same observation that helped construct it.

Bandwidth selection by data splitting

Split the data into a training sample and a validation sample:

$$X = (X_1, \dots, X_{2n}) \Rightarrow \begin{cases} Y = (Y_1, \dots, Y_n) & \text{training sample} \\ Z = (Z_1, \dots, Z_n) & \text{validation sample} \end{cases}$$

For each candidate bandwidth $h_j \in \mathcal{H}$, estimate a density using only the training data: $Y \longrightarrow \hat{p}_j = \hat{p}_{h_j}$. Then evaluate its estimated risk on the validation sample:

$$\hat{L}_j = \int \hat{p}_j^2(x) dx - \frac{2}{n} \sum_{i=1}^n \hat{p}_j(Z_i).$$

Choose the density estimator with the smallest estimated risk:

$$\hat{p} = \hat{p}_{\hat{j}}, \quad \hat{j} = \arg \min_{j=1, \dots, N} \hat{L}_j.$$

Idea: use one part of the data to construct candidate KDEs, and an independent part of the data to compare their predictive performance.

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Kernel methods: main idea

Many learning algorithms depend on the data only through inner products $\langle \mathbf{x}_i, \mathbf{x}_j \rangle$.

Given training inputs $\mathbf{x}_1, \dots, \mathbf{x}_N$, the matrix of all pairwise inner products is called the **Gram matrix**: $G_{ij} = \langle \mathbf{x}_i, \mathbf{x}_j \rangle$, $i, j = 1, \dots, N$.

Kernel methods replace these inner products by a kernel function $K(\mathbf{x}_i, \mathbf{x}_j)$, giving the **kernel Gram matrix** $K_{ij} = K(\mathbf{x}_i, \mathbf{x}_j)$.

If $K(\mathbf{x}, \mathbf{x}_j) = \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle_{\mathcal{H}}$, then the kernel Gram matrix is the Gram matrix of the transformed features $\phi(\mathbf{x}_1), \dots, \phi(\mathbf{x}_N)$.

Here $\phi(\mathbf{x})$ maps the original input $\mathbf{x}$ into a possibly high-dimensional feature space $\mathcal{H}$.

The advantage is that we can work with $K(\mathbf{x}_i, \mathbf{x}_j)$ directly, without explicitly computing $\phi(\mathbf{x}_i)$ and $\phi(\mathbf{x}_j)$. This is the **kernel trick**.

Mercer kernels

A function $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a valid kernel if it behaves like an inner product in some feature space.

Standard sufficient conditions:

- ▶ K is symmetric: $K(\mathbf{x}, \mathbf{x}') = K(\mathbf{x}', \mathbf{x})$
- ▶ K is positive semidefinite: for any $\mathbf{x}_1, \dots, \mathbf{x}_N \in \mathcal{X}$ and any $c_1, \dots, c_N \in \mathbb{R}$

$$\sum_{i=1}^N \sum_{j=1}^N c_i c_j K(x_i, x_j) \geq 0.$$

Equivalently, the Gram matrix $\mathbf{K}_{ij} = K(x_i, x_j)$ must be positive semidefinite:

$$\mathbf{c}^\top \mathbf{K} \mathbf{c} \geq 0 \quad \text{for all } \mathbf{c} \in \mathbb{R}^N.$$

Example: polynomial kernel

Consider the degree-2 polynomial kernel $K(\mathbf{x}, \mathbf{z}) = \langle \mathbf{x}, \mathbf{z} \rangle^2$.

For $\mathbf{x} = (x_1, x_2)^\top$ and $\mathbf{z} = (z_1, z_2)^\top$, $K(\mathbf{x}, \mathbf{z}) = (x_1 z_1 + x_2 z_2)^2$.

Expanding, $K(\mathbf{x}, \mathbf{z}) = x_1^2 z_1^2 + 2x_1 x_2 z_1 z_2 + x_2^2 z_2^2$.

This is an inner product in the transformed feature space

$$\phi(\mathbf{x}) = (x_1^2, \sqrt{2}x_1 x_2, x_2^2)^\top.$$

Indeed, $K(\mathbf{x}, \mathbf{z}) = \phi(\mathbf{x})^\top \phi(\mathbf{z})$.

For example, a linear model in the transformed variables θ

$$f(\mathbf{x}) = \theta^\top \phi(\mathbf{x}) = \theta_1 x_1^2 + \theta_2 \sqrt{2} x_1 x_2 + \theta_3 x_2^2.$$

is nonlinear in the original variables x_1, x_2 .

Kernel methods use $K(\mathbf{x}, \mathbf{z})$ to work with these transformed features without explicitly constructing $\phi(\mathbf{x})$.

Example: Gaussian / RBF kernel

Gaussian or radial basis function (RBF) kernel is common:

$$K(\mathbf{x}, \mathbf{z}) = \exp \left(-\frac{\|\mathbf{x} - \mathbf{z}\|^2}{2\ell^2} \right).$$

The parameter $\ell > 0$ is the length-scale, or bandwidth.

- ▶ Small ℓ : only very nearby points have high similarity.
- ▶ Large ℓ : points remain similar over longer distances.

The RBF kernel is positive semidefinite $\rightarrow$ a valid Mercer kernel.

It corresponds to an implicit feature map $\phi(\mathbf{x})$ into an infinite-dimensional feature space: $K(\mathbf{x}, \mathbf{z}) = \langle \phi(\mathbf{x}), \phi(\mathbf{z}) \rangle_{\mathcal{H}}$.

Thus kernel methods can fit nonlinear functions while avoiding explicit construction of high-dimensional features.

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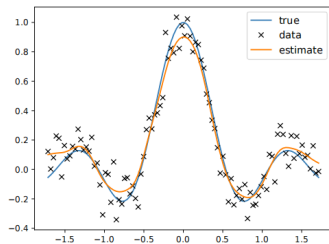
Warm-up: Nonparametric Regression

We observe $(X_1, Y_1), \dots, (X_n, Y_n)$ and we want to estimate $m(\mathbf{x}) = E(Y|\mathbf{X} = \mathbf{x})$

Idea

Use the **Nadaraya–Watson estimator** with a smoothing kernel K and a bandwidth $h > 0$ controlling the amount of smoothing (nearby observations receive larger weights).

$$\hat{m}_h(\mathbf{x}) = \frac{\sum_{i=1}^n Y_i K\left(\frac{\|\mathbf{x} - \mathbf{X}_i\|}{h}\right)}{\sum_{i=1}^n K\left(\frac{\|\mathbf{x} - \mathbf{X}_i\|}{h}\right)}.$$



Gaussian Processes: High-Level Idea

A **Gaussian process (GP)** is a distribution over functions $f : \mathcal{X} \rightarrow \mathbb{R}$, where $\mathcal{X}$ is any domain.

Key assumption

For a set of $M > 0$ inputs $\mathbf{x}_1, \dots, \mathbf{x}_M \in \mathcal{X}$, the corresponding function values $\mathbf{f} = [f(\mathbf{x}_1), \dots, f(\mathbf{x}_M)]^\top$ are assumed to be jointly Gaussian: $\mathbf{f} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where

- ▶ $\boldsymbol{\mu} = [m(\mathbf{x}_1), \dots, m(\mathbf{x}_M)]^\top$ is the **mean function**
- ▶ $\Sigma_{ij} = K(\mathbf{x}_i, \mathbf{x}_j)$ is a positive-definite **kernel**, also called the **covariance function** (controls how strongly function values at different inputs are related)

Remarks on GP

- ▶ Since the Gaussian process assumption holds for any $M > 0$, it also holds for $M = N + 1$, where we combine N training inputs with one test input $\mathbf{x}_*$.
- ▶ Therefore, we can infer the unknown value $f(\mathbf{x}_*)$ from the observed values $f(\mathbf{x}_1), \dots, f(\mathbf{x}_N)$ by conditioning the joint Gaussian distribution $p\{f(\mathbf{x}_1), \dots, f(\mathbf{x}_N), f(\mathbf{x}_*)\}$.
- ▶ The same idea extends to settings where we observe noisy versions of $f(\mathbf{x}_n)$, as in regression or classification.

Kernels in Nadaraya–Watson vs. in Gaussian processes

- ▶ Nadaraya–Watson regression uses a kernel to define **local similarity**.
- ▶ Gaussian processes uses a kernel $K(\mathbf{x}, \mathbf{z})$ to define the **covariance** between function values $f(\mathbf{x})$ and $f(\mathbf{z})$.
- ▶ **The kernel moves from being a local averaging weight to defining a full probabilistic model over functions.**

Noise-free GP observations

- ▶ Training data: $\mathcal{D} = \{(\mathbf{x}_n, y_n) : n = 1, \dots, N\}$.
- ▶ In the noise-free case, $y_n = f(\mathbf{x}_n)$.
- ▶ If the GP predicts at an input already seen in the training set, it should return $f(\mathbf{x}_n)$ with no uncertainty.
- ▶ Therefore, in the noise-free setting, the GP acts as an **interpolator** of the training data.

Prediction in noise-free case

- ▶ Let the test inputs be $\mathbf{X}_* = [\mathbf{x}_1^*, \dots, \mathbf{x}_{N_*}^*]^\top$.
- ▶ We want to predict $\mathbf{f}_* = [f(\mathbf{x}_1^*), \dots, f(\mathbf{x}_{N_*}^*)]^\top$.
- ▶ The GP defines a joint Gaussian distribution over training and test function values: $p(\mathbf{f}_X, \mathbf{f}_* \mid \mathbf{X}, \mathbf{X}_*)$.
- ▶ By definition of GP, $(\mathbf{f}_X, \mathbf{f}_*)^\top$ is jointly Gaussian:

$$\begin{pmatrix} \mathbf{f}_X \\ \mathbf{f}_* \end{pmatrix} \sim \mathcal{N} \left[\begin{pmatrix} \boldsymbol{\mu}_X \\ \boldsymbol{\mu}_* \end{pmatrix}, \begin{pmatrix} \mathbf{K}_{X,X} & \mathbf{K}_{X,*} \\ \mathbf{K}_{X,*}^\top & \mathbf{K}_{*,*} \end{pmatrix} \right].$$

- ▶ $\boldsymbol{\mu}_X = [m(\mathbf{x}_1), \dots, m(\mathbf{x}_N)]^\top$ and $\boldsymbol{\mu}_* = [m(\mathbf{x}_1^*), \dots, m(\mathbf{x}_{N_*}^*)]^\top$.
- ▶ Covariance blocks: $\mathbf{K}_{X,X} = K(\mathbf{X}, \mathbf{X})$ ($N \times N$),
 $\mathbf{K}_{X,*} = K(\mathbf{X}, \mathbf{X}_*)$ ($N \times N_*$), and $\mathbf{K}_{*,*} = K(\mathbf{X}_*, \mathbf{X}_*)$ ($N_* \times N_*$).
- ▶ These encode training–training, training–test, and test–test covariances.

Posterior predictive distribution

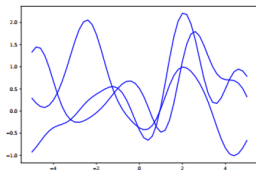
- ▶ Conditioning the joint Gaussian gives $p(\mathbf{f}_* | \mathbf{X}_*, \mathcal{D}) = \mathcal{N}(\mathbf{f}_* | \boldsymbol{\mu}_*, \boldsymbol{\Sigma}_*)$.
- ▶ The posterior mean is $\boldsymbol{\mu}_* = m(\mathbf{X}_*) + \mathbf{K}_{X,*}^\top \mathbf{K}_{X,X}^{-1}(\mathbf{f}_X - m(\mathbf{X}))$.
- ▶ The posterior covariance is $\boldsymbol{\Sigma}_* = \mathbf{K}_{*,*} - \mathbf{K}_{X,*}^\top \mathbf{K}_{X,X}^{-1} \mathbf{K}_{X,*}$.

Interpretation

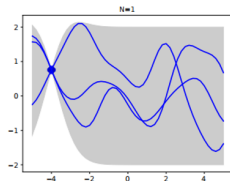
- ▶ The posterior mean $\boldsymbol{\mu}_*$ gives the GP prediction at the test inputs.
- ▶ The posterior covariance $\boldsymbol{\Sigma}_*$ gives predictive uncertainty.
- ▶ In the noise-free case, the GP interpolates the observed training points exactly.
- ▶ Predictive uncertainty is smallest near observed inputs and typically increases farther away from the training data.

GP without noise – illustration

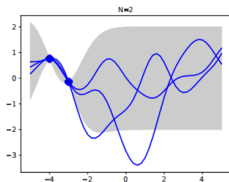
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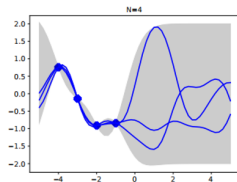
(a)



(b)



(c)



(d)

Noisy GP observations

- ▶ Now we observe noisy outputs: $y_n = f(\mathbf{x}_n) + \epsilon_n$, where $\epsilon_n \sim \mathcal{N}(0, \sigma_y^2)$.
- ▶ The GP no longer has to interpolate the data exactly.
- ▶ Instead, predictions should stay close to the observed data, with uncertainty reflecting observation noise.
- ▶ For noisy observations, $\text{Cov}[y_i, y_j] = \text{Cov}[f_i, f_j] + \text{Cov}[\epsilon_i, \epsilon_j]$.
- ▶ Hence, $\text{Cov}[y_i, y_j] = K(\mathbf{x}_i, \mathbf{x}_j) + \sigma_y^2 \delta_{ij}$, where $\delta_{ij} = \mathbb{I}(i = j)$.
- ▶ In matrix form, $\text{Cov}[\mathbf{y} \mid \mathbf{X}] = \mathbf{K}_{X,X} + \sigma_y^2 \mathbf{I}_N \triangleq \mathbf{K}_\sigma$.

Joint and predictive distributions

Joint distribution with noisy training data

- ▶ We condition on the noisy observations $\mathbf{y}$, but predict the latent noise-free test values $\mathbf{f}_*$.
- ▶ The joint distribution is

$$\begin{pmatrix} \mathbf{y} \\ \mathbf{f}_* \end{pmatrix} \sim \mathcal{N} \left[\begin{pmatrix} \boldsymbol{\mu}_X \\ \boldsymbol{\mu}_* \end{pmatrix}, \begin{pmatrix} \mathbf{K}_\sigma & \mathbf{K}_{X,*} \\ \mathbf{K}_{X,*}^\top & \mathbf{K}_{*,*} \end{pmatrix} \right].$$

Posterior predictive distribution

- ▶ Conditioning the joint Gaussian gives $p(\mathbf{f}_* \mid \mathcal{D}, \mathbf{X}_*) = \mathcal{N}(\mathbf{f}_* \mid \boldsymbol{\mu}_{*|X}, \boldsymbol{\Sigma}_{*|X})$.
- ▶ The posterior mean is $\boldsymbol{\mu}_{*|X} = \boldsymbol{\mu}_* + \mathbf{K}_{X,*}^\top \mathbf{K}_\sigma^{-1}(\mathbf{y} - \boldsymbol{\mu}_X)$.
- ▶ The posterior covariance is $\boldsymbol{\Sigma}_{*|X} = \mathbf{K}_{*,*} - \mathbf{K}_{X,*}^\top \mathbf{K}_\sigma^{-1} \mathbf{K}_{X,*}$.

Single test input

- ▶ For one test input $\mathbf{x}_*$, define $\mathbf{k}_* = [K(\mathbf{x}_*, \mathbf{x}_1), \dots, K(\mathbf{x}_*, \mathbf{x}_N)]^\top$ and $k_{**} = K(\mathbf{x}_*, \mathbf{x}_*)$.
- ▶ Then $p(f_* \mid \mathcal{D}, \mathbf{x}_*) = \mathcal{N}(f_* \mid m_*, k_{**} - \mathbf{k}_*^\top \mathbf{K}_\sigma^{-1} \mathbf{k}_*)$.
- ▶ The mean is $m_* + \mathbf{k}_*^\top \mathbf{K}_\sigma^{-1} (\mathbf{y} - \mu_X)$.

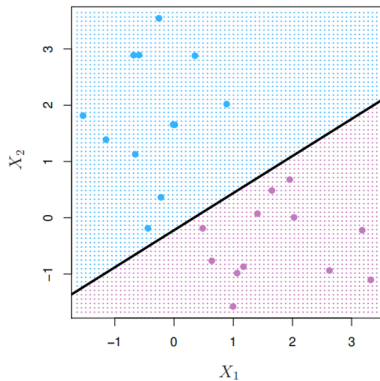
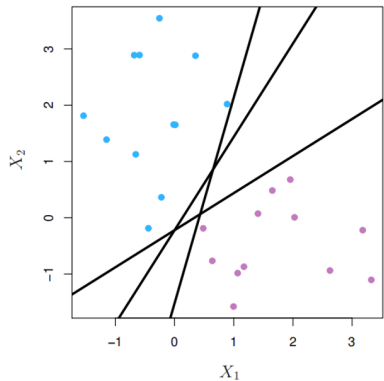
Zero-mean case and kernel ridge regression

- ▶ If $m(\cdot) = 0$, the posterior mean simplifies to $\mu_{*|X} = \mathbf{k}_*^\top \mathbf{K}_\sigma^{-1} \mathbf{y}$.
- ▶ Equivalently, $\mu_{*|X} = \sum_{n=1}^N K(\mathbf{x}_*, \mathbf{x}_n) \alpha_n$, where $\boldsymbol{\alpha} = \mathbf{K}_\sigma^{-1} \mathbf{y}$.
- ▶ This has the same form as predictions from **kernel ridge regression**.

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- ▶ Exemplar methods
 - ▶ k-nearest neighbor (kNN)
 - ▶ kernel density estimation (KDE)
- ▶ Kernel methods
 - ▶ Mercer Kernels
 - ▶ Gaussian process (GP)
 - ▶ Support Vector Machine (SVM)

What is a “good” classifier?



Hyperplane

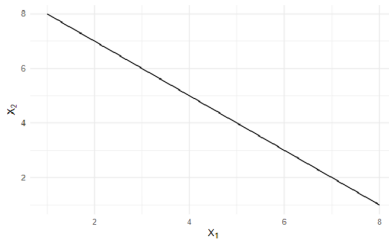
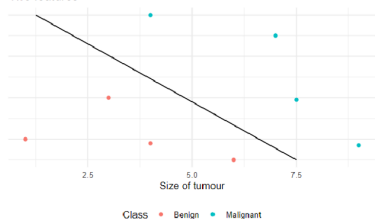
- ▶ In p dimensions, it is a flat affine subspace of dimension $p - 1$
- ▶ General equation has the form:

$$\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p = 0$$

- ▶ In $p = 2$ dimensions, the hyperplane is a line
- ▶ If $\beta_0 = 0$, the hyperplane passes through the origin, otherwise it does not.
- ▶ The vector $(\beta_0, \dots, \beta_p)$ is called the normal vector $\rightarrow$ it points to a direction orthogonal to the surface of the hyperplane.

Hyperplane in $p = 2$

Two features



Equation of the hyperplane here is $-9 + X_1 + X_2 = 0$

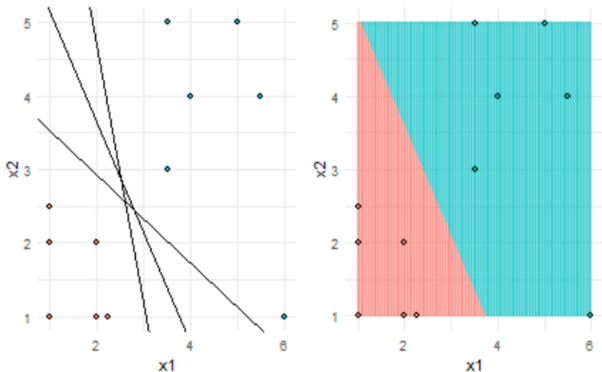
Separating hyperplanes

Consider class 1 (red) as $Y_i = 1$ and class 2 (blue) as $Y_i = -1$.

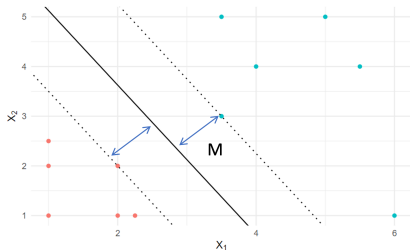
Then $y_i f(x_i) > 0$ for all i ; $f(x_i)$ defines a separating hyperplane.

If $f(X) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$ defines a hyperplane:

- ▶ $f(x) > 0$ defines a region on one side of the hyperplane,
- ▶ $f(x) < 0$ on the other side.



Maximal Margin Classifier



$$\max_{\beta_0, \beta_1, \beta_2, \dots, \beta_p} M \quad \text{such that} \quad \sum_{j=1}^p \beta_j^2 = 1$$
$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}) \geq M$$

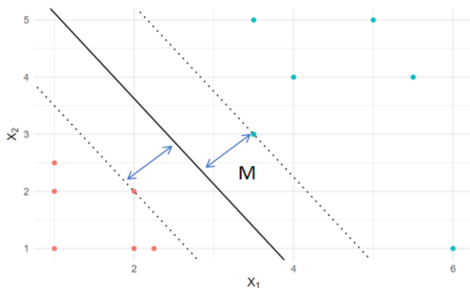
- ▶ Given a hyperplane, what is the distance between a point to the hyperplane?
- ▶ What is the “optimal” way to separate samples from two classes?
- ▶ Which part of samples are more “important” to determine the hyperplane?

Maximal Margin Classifier

After some re-formulation and reparameterization, we have

$$\min_{\beta_0, \beta_1, \dots, \beta_p} \frac{1}{2} \|\beta\|^2$$

such that $y_i(x_i\beta + \beta_0) \geq 1$



quadratic programming!

From the primal to the dual

The hard-margin SVM starts with the primal problem:

$$\min_{\beta_0, \beta_1, \dots, \beta_p} \frac{1}{2} \|\beta\|^2$$

such that $y_i(x_i\beta + \beta_0) \geq 1$

Interpretation:

- ▶ We want a separating hyperplane.
- ▶ We want the margin to be as large as possible.
- ▶ Every training point must be on the correct side of the margin.

The dual version rewrites the problem using one weight per training point: $\alpha_i \geq 0$.

Think of α_i as the **importance weight** of observation i .

The key Karush-Kuhn-Tucker (KKT) idea

The important KKT condition is: $\alpha_i (y_i f(x_i) - 1) = 0$. This means one of the two terms must be zero.

Case 1: Point is far from the margin: $y_i f(x_i) > 1$

Then $y_i f(x_i) - 1 > 0$ so we must have: $\alpha_i = 0$.

This point does **not** affect the final classifier.

Case 2: Point is exactly on the margin: $y_i f(x_i) = 1$

Then α_i may be positive.

These points are the **support vectors**.

Support vectors

The final SVM classifier can be written as:

$$f(x) = \beta_0 + \sum_{i=1}^n \alpha_i y_i x_i^\top x.$$

But most α_i 's are zero. So only points with $\alpha_i > 0$ actually contribute to the classifier.

These points are called **support vectors**.

Main takeaway:

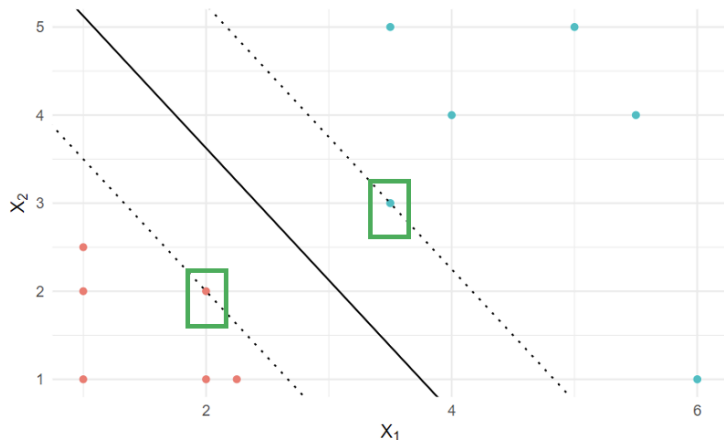
The SVM boundary is determined only by the support vectors.

Points far away from the margin do not matter, because their $\alpha_i = 0$.

Illustration

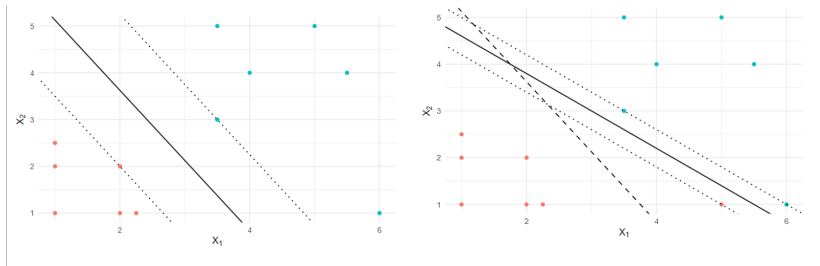
Only $\alpha_i > 0$ contribute to final classification function

Only $(y_i f(x_i) - 1) = 0$ could lead to $\alpha_i > 0$



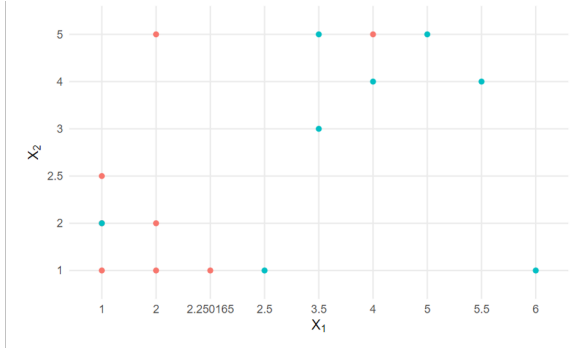
Support vector/support set

Effect of noisy data



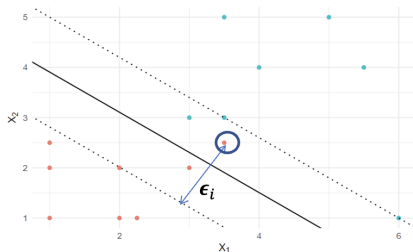
- ▶ Data could be separable, but noisy $\rightarrow$ unstable solution for the maximal margin classifier.
- ▶ The support vector classifier maximizes a soft margin.
 - ▶ relaxes requirement for all observations to be on the correct side of the margin.

Non-perfect separation



- ▶ There is no linear boundary (hyperplane) that perfectly separates the classes
- ▶ This is typically the case that observations don't have a perfect boundary of separation
 - ▶ relaxes requirement for **all observations** to be on the correct side of the margin.

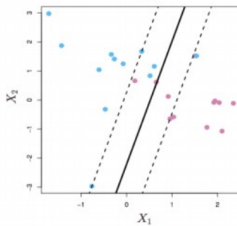
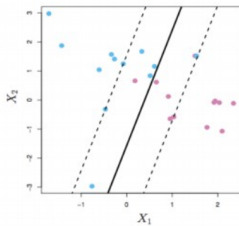
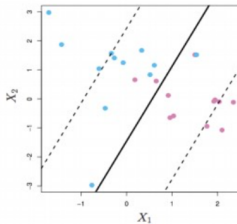
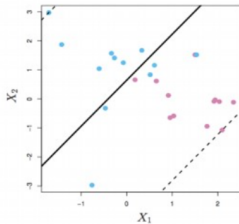
Slack variables in support vector classifier



$$\begin{aligned} \max_{\beta_0, \beta_1, \beta_2, \dots, \beta_p, \epsilon_1, \dots, \epsilon_n} \quad & M \quad \text{such that} \quad \sum_{j=1}^p \beta_j^2 = 1 \\ y_i (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}) &\geq M(1 - \epsilon_i) \\ \epsilon_i &\geq 0, \quad \sum_{i=1}^n \epsilon_i \leq C \end{aligned}$$

Impact of parameters C

Largest C

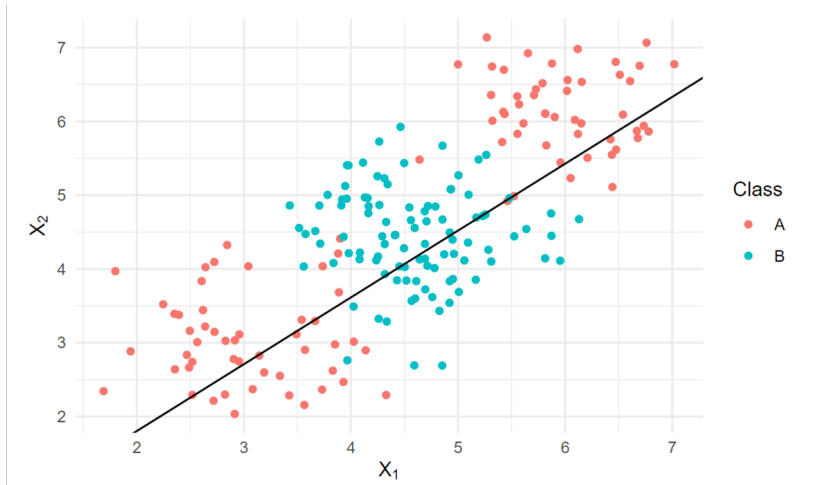


Smallest C

Quick summary

- ▶ Hyperplane
- ▶ Maximal margin classifier
- ▶ Support vector classifier

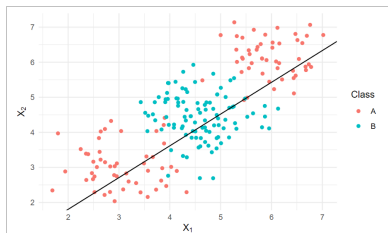
How about the following case?



► Single linear boundary can be insufficient

Feature space expansion

- For the data points not (linear) separable in lower dimensional space, if we map (denote as ϕ) them into higher dimensional space, it is more likely to be separable. (VC dimension!)



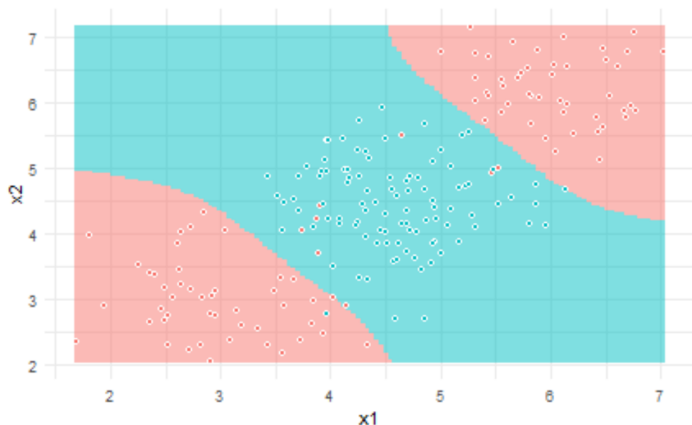
$$\phi : (X_1, X_2) \rightarrow (X_1, X_2, X_1^2, X_2^2, X_1X_2)$$

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$$

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1^2 + \beta_4 X_2^2 + \beta_5 X_1 X_2 = 0$$

Feature space expansion

- Using degree six polynomial expansion



- In general, after mapping, decision boundary can be written as:

$$\{x \in \mathbb{R}^p : \beta_0 + \phi(x)\beta = 0\}$$

Kernel tricks

- ▶ Recall inner product:
- ▶ Let $x_i = (x_{i1}, \dots, x_{ip})$, the inner product between vectors given by $\langle x_i, x_j \rangle = \sum_{k=1}^p x_{ik} x_{jk}$
- ▶ The linear support vector classifier can be represented as (duality)

$$f(x) = \beta_0 + \sum_{i=1}^N \alpha_i \langle x, x_i \rangle$$

$$f(x) = \beta_0 + \sum_{i=1}^N \alpha_i \underbrace{\langle \phi(x), \phi(x_i) \rangle}_{K(x, x_i)}$$

Support vector machine

- ▶ The support vector machine (SVM) is an extension of the support vector classifier that obtains non-linear decision boundaries by (implicitly) enlarging the feature space using kernels.

$$f(x) = \beta_0 + \sum_{i=1}^N \alpha_i K(x, x_i)$$

- ▶ As before, the classifier will be based on a hyperplane (in the higher dimensional space). The classifier will have the form:

$$\hat{y} = \text{sign}(f(x))$$

Some kernels

- Polynomial kernel

$$K(x_i, x_j) = (1 + \langle x_i, x_j \rangle)^d$$

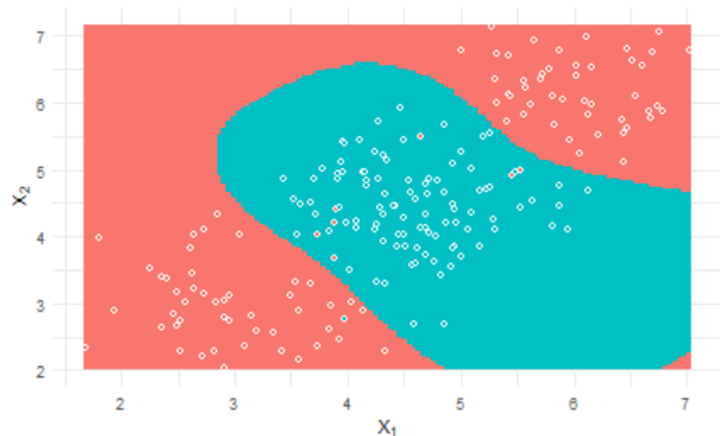
- Gaussian radial kernel/Radial basis function kernel

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

- Gaussian radial kernel (1d) as expansion of feature mapping:

$$\phi(x) = e^{-\gamma x^2} \left[1, \sqrt{\frac{\gamma}{1!}} x, \sqrt{\frac{\gamma^2}{2!}} x^2, \dots, \sqrt{\frac{\gamma^n}{n!}} x^n, \dots \right]$$

Support vector machines with the radial kernel



SVM for multiclass

- ▶ SVM covered previously are design for binary classification. To expand this to K classes there are two options:
 1. One vs one:
Fit all $\binom{K}{2}$ pairwise classifiers. Then, classify x to the class that wins the most pairwise competitions.
 2. One vs all:
Fit k different 2-class SVM classifiers; each class versus the rest. Classify x to the class for which the signed distance to the decision boundary is the largest.
- ▶ Which to use?

If k is small, do one vs one. Otherwise recommended one vs all.

References

- ▶ [ESLII](#): Chapters 3.1, 4.1-4.4
- ▶ [PRML](#) Chapters 3.1, 4.1
- ▶ [PML](#): Chapter 16-17